

Explainable Multi-Modal Stock Market Forecasting Using Dynamic Graph Transformers, Financial Large Language Models, and Retrieval-Augmented Generation

Abstract :

Accurate stock market forecasting remains one of the most challenging problems in financial intelligence due to market volatility, complex inter-stock dependencies, rapidly evolving economic conditions, and the influence of unstructured textual information. Traditional forecasting approaches primarily rely on historical price movements and technical indicators, limiting their ability to incorporate real-time financial knowledge and provide interpretable predictions. This paper proposes an **Explainable Multi-Modal Stock Market Forecasting Framework** that integrates **Dynamic Graph Transformers**, **Financial Large Language Models (LLMs)**, **Retrieval-Augmented Generation (RAG)**, and **Explainable Artificial Intelligence (XAI)** to improve prediction accuracy and transparency.

The proposed framework models the stock market as a dynamic graph in which stocks are represented as nodes and evolving relationships are learned from sector information, correlation structures, and market behavior. Historical market data, technical indicators, and temporal patterns are processed through a Dynamic Graph Transformer to capture both spatial and temporal dependencies. Simultaneously, a Financial LLM is enhanced with a Retrieval-Augmented Generation pipeline that retrieves relevant information from financial news, SEC filings, earnings reports, analyst reports, macroeconomic indicators, and social media, enabling the model to reason over current market events instead of relying solely on parametric knowledge. The numerical and textual representations are fused through an adaptive attention mechanism to generate robust price forecasts.

To improve model transparency and investor trust, Explainable AI techniques including **SHAP** and **attention visualization** are incorporated to identify the most influential market features, retrieved financial documents, and graph relationships contributing to each prediction. Unlike conventional black-box forecasting systems, the proposed framework produces both quantitative predictions and human-interpretable explanations that support financial decision-making.

Experimental evaluation is conducted on historical stock market data augmented with real-time financial documents and sentiment information. Performance is assessed using forecasting metrics including **MAE**, **RMSE**, **MAPE**, and **Directional Accuracy**, along with interpretability analysis. Results demonstrate that integrating Dynamic Graph Transformers with Financial LLMs, Retrieval-Augmented Generation, and Explainable AI significantly enhances predictive performance while providing transparent and trustworthy investment insights suitable for intelligent financial decision-support systems.

Keywords :

Stock Market Prediction, Dynamic Graph Transformer, Financial Large Language Models, Retrieval-Augmented Generation, Explainable Artificial Intelligence, Graph Neural Networks, Time-Series Forecasting, Multi-Modal Learning, Financial Intelligence, Deep Learning.

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